

Machine Learning (IE 406)

Course Placement: It is a technical elective course for B. Tech. 5th and 7th Semester students

Course format: Three hours' lecture and two-hours lab every week (3-0-2-4).

Course Content:

S. No.	Unit Title	Topics	Lectures
1	Introduction to ML	Introduction and Significance of Machine Learning, Supervised, Semi-supervised and Unsupervised learning, Introduction to Classification, Regression and Clustering problems.	2
2	Mathematical and Statistical Foundation	Quick revision of linear algebra for machine learning, Convex functions, Gradient, Steepest descent direction, Gradient Descent method, Stochastic Gradient Descent method, Sub gradients.	6
		Multivariate Analysis , Covariance matrix, Multivariate normal distribution and its properties,.	
3	Model Selection and Evaluation	Evaluation metric for regression and classification models, Training and Testing set, Resampling method, Cross Validation, K-fold cross validation, Leave-one Out method.	2
4	Regression Analysis	Least Squares methods, Least Squares multiple linear regression model, Non-linear multiple regression models through basis functions, Polynomial, Gaussian and Sigmoidal basis, Overfitting, Least Squares kernel regression model, Derivation of least square loss for normal noise using MLE, Need of Regularization, Bias-variance trade-off, Lasso Regression, L_1 - norm loss regression model,	11
4.	Classification	Bayesian Classification, Gaussian Models, Naïve Bayes Classifier, Logistic Regression, Soft-max Regression, Neural Network and multilayer perceptron and Support Vector Machine.	14
6.	Unsupervised ML	Feature Selection methods, Dimensionality Reduction: PCA, Basic of clustering, K-mean clustering.	5

References:

Suggested books:

For development of ML general concepts: -

- Bishop, Christopher M. *Pattern recognition and Machine learning* 128.9 (2008).
 - A very popular and standard book for developing the ML general concepts systematically.
- Murphy, K. P. (2013). *Machine learning: a probabilistic perspective*. Cambridge, Mass: MIT Press. ISBN: 9780262018029 026201802.
 - A detailed and widely adapted book for providing the statistical perspective of ML general concepts.
- Friedman, Jerome, Trevor Hastie, and Robert Tibshirani. *The elements of statistical learning*. Vol. 1. No. 10. New York: Springer series in statistics, 2001.
 - It has nice stories which avoid the mathematical complexity and comprehend the ML concepts easily.
- Duda, Richard O., and Peter E. Hart. *Pattern classification*. John Wiley & Sons, 2006.
 - A popular book of ML particularly for the Bayesian learning.
- Zaki, Mohammed J., and Wagner Meira Jr. *Data Mining and Machine Learning: Fundamental Concepts and Algorithms*. Cambridge University Press, 2020.
 - A good reference book for overall ML techniques and algorithms.
- Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong. *Mathematics for machine learning*. Cambridge University Press, 2020.
 - Use it to understand the mathematics required for building the ML concepts.
- Chandra, Suresh, Jayadeva, and Aparna Mehra. *Numerical optimization with applications*. Alpha Science International, 2009.
 - Use it to understand the optimization methods used for building the ML concepts.
- Deng, Naiyang, Yingjie Tian, and Chunhua Zhang. *Support vector machines: optimization based theory, algorithms, and extensions*. CRC press, 2012.
 - It provides the SVM comprehension from the lens of optimization.
- Zaki, Mohammed J., and Wagner Meira Jr. *Data Mining and Machine Learning: Fundamental Concepts and Algorithms*. Cambridge University Press, 2020.

Evaluation Components: - Surprise Quizzes, Assignments, Viva, Semester exams, Lab test.

Tentative marks distributions: -

- Surprise Quizzes:- 30 .
- End Semester Exam:- 30.
- Lab Assignments +Viva + Lab test:- 40.

Evaluation are subject to change as per the response of the class.

Course Outcomes: After completion of this course, students will

- develop the depth understanding of the fundamental mathematical and statistical concepts used in machine learning models.
- develop the systematic and deep understanding of the popular machine learning models.
- learn to implement the machine learning models from scratch.
- will be able to apply the variants of machine learning model to solve the real-time problems.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X			X	X							